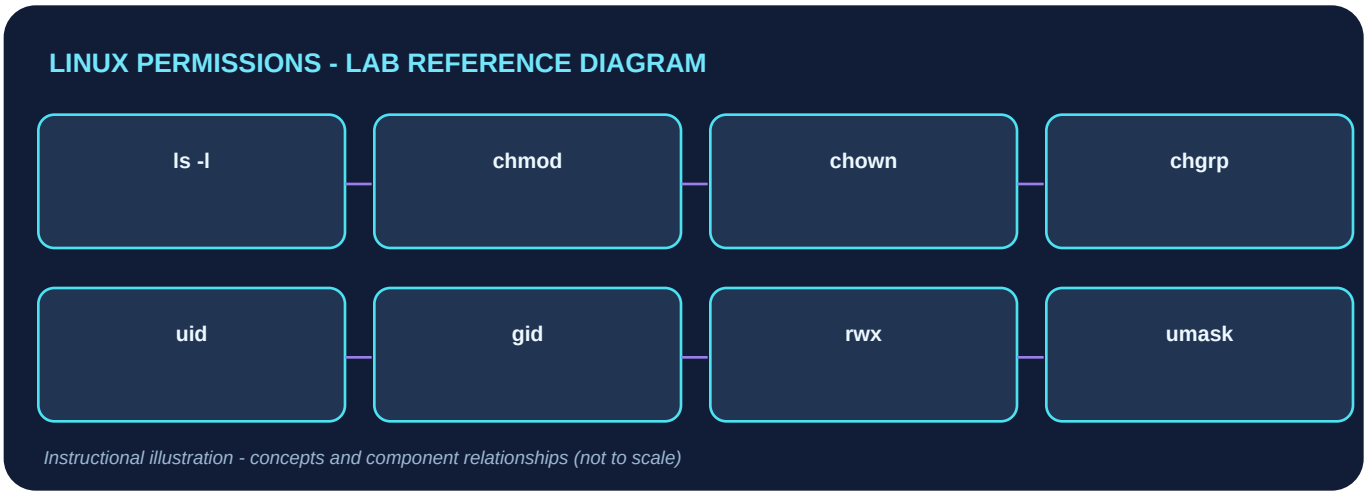
17.1.10 Manage Linux File Ownership

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Work with chown, chgrp and chmod.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Read ownership and mode using ls -l.
2. Calculate permissions using r=4, w=2, x=1.
3. Apply a safe mode such as chmod 640 to practice files.
4. Verify user/group access without loosening security.

Troubleshooting and safe practice

• Permission denied: check parent directory execute bit.

• Wrong owner: use authorized chown or chgrp.

17.1.10 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
ls -l	A key term in linux permissions; identify its purpose, where it is used, and how to verify it.
chmod	Change file or directory permission bits.
chown	Change a file or directory owner.
chgrp	Change a file or directory group.
uid	Linux access-control value used to determine who may access a resource.
gid	Linux access-control value used to determine who may access a resource.
rwx	A key term in linux permissions; identify its purpose, where it is used, and how to verify it.
umask	A key term in linux permissions; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____